COM 480 - Data Visualization

Milestone 2

École Polytechnique Fédérale de Lausanne

AquaViz

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1 Project idea

1.1 Main idea

For the core visualisation, the aim would be to create an interactive map where each country is represented by a colour indicating the level of water scarcity. By clicking on a country, users could view various water consumption statistics: a doughnut chart showing different consumption levels, per capita consumption, etc. In addition, there would be a bar at the top of the page where users could select the year for which they wish to view the statistics.

1.2 Challenges and extensions

Extension 1
An interesting extension of this project would be to add a trend line for the selected country. As the data covers the period from 2000 to 2024, it would be possible to display a visual representation of the trend in water consumption.
Extension 2
Furthermore, it would be useful to display a time-series animation of the map showing the level of scarcity by country, so as to observe changes over time. Moreover, the time-series animation could also provide predictions of the level of water scarcity for the future.
Extension 3
This dataset contains data from 20 countries, so an interesting visualization would be to select two countries and observe their different trajectories.
Extension 4
Another interesting extension would be to highlight key historical moments that have shaped water consumption in certain countries, such as the water crisis in Cape Town [1].

2 Tools and Methods

- **Tools & Deployment:**
  - **HTML5, CSS, and JavaScript (ES6):** These core technologies will be used to build the DOM skeleton, manage the layout, and handle application logic.
  - **D3.js:** We will use D3.js for our Interactive Geographical Map.
  - **GeoJson:** To power the interactive timeline and dynamically filter the dataset by year across all views efficiently.
  - **Deployment Strategy (Flask & GitHub Pages):** For this MVP, we kept it simple with a static HTML/JS setup. Moving forward, we will introduce a local Flask server to handle the data. For GitHub Pages, which requires static files, we'll build a script that exports our dynamic project into ready-to-use HTML pages for the final deployment.
- **Techniques Lecture:**
  - **Maps:** Our core visualization is a Choropleth map, using GeoJSON and mapping tools (D3 Geo or Leaflet) to render geographical polygons.

– **Perception:** For the map, we will use country polygons (Marks) and choose an accessible color scheme (Channel) to accurately represent water scarcity without misleading the viewer.
- **Interactions:** Timeline sliders and map clicks will dynamically update the side charts (linked composite views), allowing users to explore data efficiently without cognitive overload.
- **Storytelling:** Extension 4 introduces narrative elements to guide the user through impactful historical water crises (e.g., Cape Town), adding context to the raw data.

3 Implementation Plan

We have broken down our goals into pieces:

- **Phase 1 (Minimum Viable Product Base):** Prepare and clean the dataset, define the mapping between country names and GeoJSON features, and build the basic HTML/CSS skeleton.
- **Phase 2 (Core Visualization):** Implement the interactive map, timeline (year selection) slider, and linked side charts. The core dashboard will allow users to select a year, click on a country, and view key indicators such as total water consumption, water scarcity level, and per capita water use.
- **Phase 3 (Extensions & Deployment):** Add time-series animations, historical event annotations, and other advanced features. Finally, deploy the final static version of the project to GitHub Pages.

4 Sketch and Skeleton website

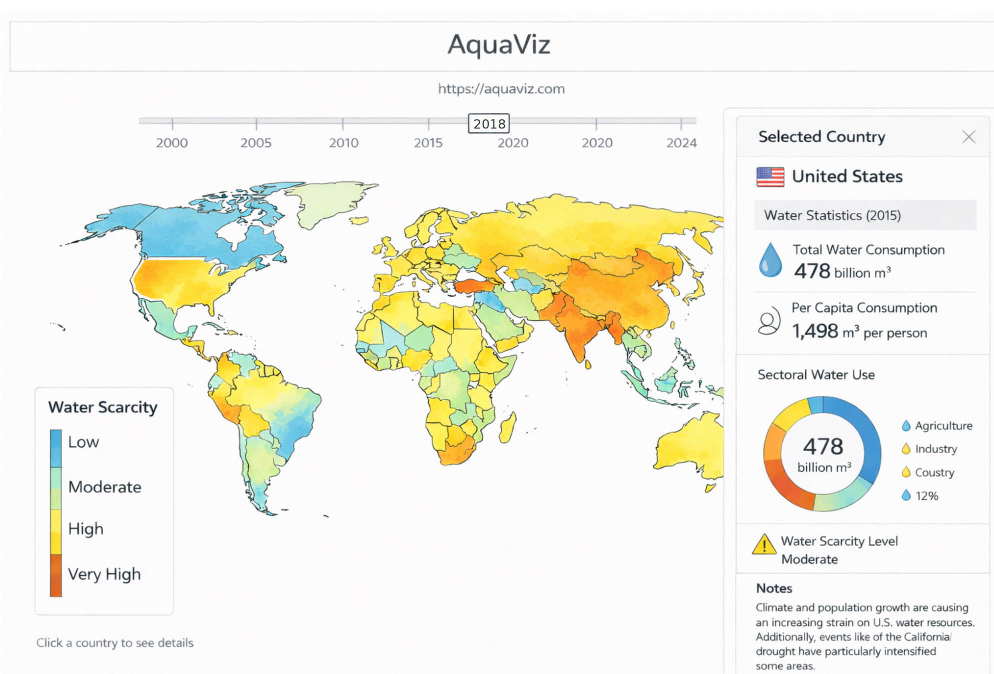

Figure 1: Sketch of the Website [2]

Here the link of the skeleton website : Website AquaViz

References

[1] National Geographic Society. *How Cape Town Avoided Day Zero, and What We Can Learn*. Last updated June 24, 2025. June 2025. URL: https://education.nationalgeographic.org/resource/why-cape-town-running-out-water-and-whos-next/ (visited on 04/03/2026).

[2] OpenAI. *ChatGPT*. https://openai.com/index/chatgpt/. Accessed: 2026-04-13. 2022.